

SPECTRAVIDEO

USERS GROUP -- Wgtn

Volume 3, Number 5 -- March/April 1987

N O A P R I L M E E T I N G

The next meeting is not until Monday, May 18. The Meeting which would have been held on Monday 20 April has been cancelled as this is a holiday - HAPPY EASTER FOLKS !!!!!!!

N O A P R I L M E E T I N G

I have thought that an informal get together might be nice - and the following Monday 27 April 1987 seems a good day for it. It could take the form of an evening get-together over coffee (ie bring a plate for a small supper. Phone me if interested and I will tell you the venue. Steve Hanson Ph: 729459 x8748 (w)

Wellington SV Users Group

Committee Members

Chairman ----- Darryl Alison Ph 85522 (Paraparaumu)

Secretary ----- Joseph Dol 889275

Treasurer ----- Ross Fletcher 788454

Don Stanley 896379

Garry Clark 367872

Scottie Baxter

Newsletter Committee

Steve Hanson 843495

Joseph Dol

Other specific areas of responsibility for committee members are

Don -- Software Consultant

Jim Oliver -- Radio Access & public domain tape software;

Scottie -- Library;

Darryl -- Program evaluation committee chairman & Kapiti users representative.

Ross -- Public Domain CP/M

NOTE - OUR VENUE IS NOW AT NEWLANDS INTERMEDIATE - SEE YOU THERE
- - -> FOR OUR *** MAY *** MEETING ON 18 MAY 1987 <- - - - -

GAMES SCORES

Spectron	12910	Robert Campin
Sector Alpha	2445200	Garry Clark
Kung Fu	82435	Garry Clark
	48705	Garry Clark (*)
S C Force	1640000	Alexander Lindsay
Telebunny	113520	Garry Clark
	32720	Garry Clark (+)
	31670	Kalima Collinson (*)
Omac	35360	Richard Bremford
	25700	Don Stanley (+)
Ninja	251350	Nicola Tuinman
Tetra	579650	Scottie Baxter
Turboat	68390	Garth Thornton (*)
Turboat	104610	Taane Clark
Sasa	172950	Steve Lacey
Frantic Fred	21720	Rachel Hughes (level 4)
Flipper Slip	37640	Garth Thornton
Bats Alive	3540	Colin Chin (*)
Galaxy	1724	Scottie Baxter
Jumping Jack	2600	Colin Chin (*)
TV Nightmare	7040	-
	3640	- (*)
	4980	Deirdre Chin (+)

(+) = started in normal level

(*) = started in arcade level

otherwise practice level or unknown.

A V O T E O F T H A N K S F R O M T H E E D I T O R

Without the help of a FEW club members, this bulletin would be a mere shadow .. To those members - Jim Oliver, Graeme Dick, Joe Dol, Ross Fletcher, Scottie Baxter, Don Stanley - who have helped me by supplying articles for recent issues, T H A N K Y O U !

Although this list is not exhaustive, there are a lot of you out there who are writing programs that others would be interested in, try sending an article or two in to me care of the Club P.O. Box:- Programs - Basic, Pascal, Assembly, You Name It.
 - Programming hints, and techniques.
 - Utilities - any type, as long as they have instructions.
 - News - anything other club members might like to know.
 - Problems - we will find someone to answer them if we can't.

Everything we receive will be considered. But please make sure the printing on your article is CLEAR with DARK PRINT - PLENTY of space around the Borders too! as the newsletter goes through several copying processes before printing. For Best results - send your article on a disk or tape - we WILL return it to you !!

Public Domain CP/M

Anyone wanting anything from the library of CP/M PD software, please write to Ross Fletcher at the club address, specifying either which disc, or which programs you would like. The listings in previous issues are printed from double sided disks therefore as most drives are single sided, a whole disc from the list now occupies up to 2 full discs. Most of the programs on the discs are 'Squeezed', this means they have been run through a function in New Sweep (NSW, Nsweep etc. but not the earlier Sweep) so as to take up as little room as possible. They all have a 'Q' as the 2nd letter in their extension, and can be easily unsqueezed by using New Sweep to tag them and then the 'Q' option to unsqueeze them, usually onto another disc. Please send discs, rather than asking us to purchase them for you, and please send them formatted. Please also include, either a stamped return mailer, or add 60c to your payment to cover postage. We will return your disk in your packaging.

P L E A S E R E M E M B E R Y O U R

S U B S C R I P T I O N

A plea from your committee!

There are a number of outstanding subscriptions dating back over two months in some cases.

As it is your subscriptions which keep the club running, if you want the support of the Group to continue, please help us by checking your subscription renewal date.

Your subscription renewal date is printed on the address label of every newsletter.

As the Group does not send out any other reminders for subscriptions, we have been allowing some latitude with issues being produced every two months.

If your subs are due before say May, please forward them to the Club's P.O. Box 26050 - Newlands, WELLINGTON.

Remember - the subs are only;

\$18.00 For NZ Residents
NZ\$30 For Australasian Subscriptions
NZ\$40 For other areas overseas.

Another Plea

RADIO ACCESS

BROADCASTS

You may have noticed that the Broadcasts have resumed, with February's program tape being run on the first Saturday of April. So keep an eye on the Listener - which will tell you when the next one is on ... ie the first Saturday of each month - before normal transmission commences.

So please keep contributions coming in to Jim ...
If anyone has a lack of Inspiration how about some

Holiday-time Programs

for the May Radio Access Broadcasts.

A SIMPLE SCREEN EDITOR

The following listing was originally printed as part of The Exploring Basic Series (Part 3) which I did not have time to type.

The explanation is as follows:

Just position the cursor using the cursor controls, and type in the desired characters. Key 10 can be used to switch from commands to characters. Key 5 will show the current uses of Keys 1 to 9. The key prompts are turned off and most keys nulled to reduce erroneous input. To save a screen, type in a file name and (after setting the cassette properly if using one) press the save Key. Note, the cursor is turned on during the save, so position it correctly before saving. This program uses a bug in the SPRITE\$ command to enable input without disrupting the screen (the explanation of the bug appeared in our last issue :- Steve).

LISTING ONE.

```
1 ' Listing 1    Exploring BASIC Pt3
2 '    A 40 column screen editor  Saves text as screen save files
3 ' Run, then press F5 for current Key allocations. then press
  <Enter> You may then press F10 to swap amongst 3 sets of F Keys
4 ' DO NOT UNDER ANY CIRCUMSTANCES PRESS TWO F KEYS IN SUCCESSION
  Display MENU (F5).
  Select the function you want.
  Press <enter> to clear F Key Prompt
  THEN press the selected F Key
```

```

5 ' If the option you require is not displayed, Press F10.
   This will swap to the next set of function key definitions.
   Press F5 will display these. Remember to press <Enter>
   key after F5 to clear the prompts,
6 ' When the key prompts are off, text may be entered onto the
   screen. NOTE This does not create an ASCII text file.
7 ' Command List: (Note in this listing that the characters
   '>','<','^','v'; are substituted for the arrow characters ,
   ASCII codes 212,213,214,215)
8 ' LNINPUT = line input to DD$ -(NB:- DD$ is not used )
   CLS-EoS = Clear to end of screen
   Scroll# = Scroll in direction # -from last cursor position
   SWAP = swap fkeys from commands to the arrow characters &c
9 ' C.OFF/C.ON = Cursor off/on
   INVERSE/NORMAL = INV/NORM characters (Steve Hanson Editor)

10 WIDTH 40
20 MOTOROFF: SCREEN1,3: SCREEN 0,0: CLEAR 2000: ON ERROR GOTO 430:
STOP ON: ON STOP GOSUB 430: DIM
A$(32): KF=1: DV$="DISK": C$=CHR$(27): O$=C$+"x5": Z$=C$+"y5": S$=C$+"p
": E$=C$+"q": GOSUB 240
21 'If you encounter an error when inputting the program, delete
   the error and stop trapping from the above line. Change DV$ to
   "DISK" or "CAS" as appropriate
30 JX$=INKEY$: PRINT JX$;
40 GOTO 30
50 OP$="SAVE": GOSUB 390: IF F$="" THEN RETURN
60 GOSUB 360: IF DV$="CAS" OR LEFT$(F$,3)="CAS" THEN GOSUB 440:
CSAVE F$,S ELSE IF DV$="DISK" THEN SAVE F$,S
70 GOTO 100
80 OP$="LOAD": GOSUB 390: IF F$="" THEN RETURN
90 GOSUB 360: IF DV$="CAS" OR LEFT$(F$,3)="CAS" THEN GOSUB 44:
CLOAD F$ ELSE IF DV$="DISK" THEN LOAD F$
100 GOSUB 370: RETURN
110 GOSUB 360: LOCATE 0,0: PRINT C$*M";: GOSUB 370: RETURN
120 GOSUB 360: LOCATE 0,0: PRINT C$*L";: GOSUB 370: RETURN
130 RETURN
140 GOSUB 300: GOSUB 320: ON KF GOSUB 160,170,180: GOSUB 420
150 GOSUB 310: RETURN
160 LOCATE 0,1: PRINT "SAVE LOAD Scroll^ Scrollv Menu";: LOC
ATE 0,3: PRINT "LNINPUT CLS-EoS Scroll> Scroll< SWAP";: RETURN
170 LOCATE 1,1: PRINT "> < ^ v Menu";: LOCA
TE 0,3: PRINT "INVERSE NORMAL C.OFF C.ON SWAP";: RETURN
180 LOCATE 1,1: PRINT "J . S u Menu";: LOCA
TE 0,3: PRINT "z u k i SWAP";: RETURN
190 LINEINPUT DD$: RETURN
200 PRINT C$*J";: RETURN
210 GOSUB 360: GOSUB 330: GOSUB 350: OS = 0 : GOSUB 380:
GOSUB 370: RETURN
220 GOSUB 360: GOSUB 340: GOSUB 350: OS = 39: GOSUB 380:
GOSUB 370: RETURN
230 RETURN

```

```

240 KEY 5,"": KEY 10,""
250 IF KF = 1 THEN FOR A = 1 TO 4: KEY A,CHR$(211+A): NEXT:
KEY 6,S$: KEY 7, E$: KEY 8, O$: KEY 9, Z$: KN = 2
260 IF KF = 2 THEN FOR A = 1 TO 4: KEY A,CHR$(215+A):
KEY A+5,CHR$(219+A):NEXT:KN=3
270 IF KF = 3 THEN FOR A = 1 TO 4: KEY A,"": KEY A+5,"":NEXT:KN=1
280 KEY ON: IF KN<>1 THEN KEY OFF
290 KEY(5) ON: KEY(10) ON: ON KEY GOSUB 50,80,110,120,140,190,200
,210,220,250: KF = KN: RETURN
300 GOSUB 360: FOR A = 5 TO 1 STEP -1: A$(A) = RIGHT$(SPRITE$
(63+A),31) +LEFT$(SPRITE$(64+A),1): SPRITE$(63+A) = STRING$(32,0)
:NEXT:RETURN
310 FOR A = 1 TO 5: SPRITE$(63+A)=A$(A): NEXT: GOSUB 370: RETURN
320 PRINT$;:FOR A = 1 TO 5: FOR B = 0 TO 1: LOCATE 8*(A-1),B*2:
PRINT USING"KEY###";A+B*5;: NEXT B,A: PRINT E$;:RETURN
330 FOR A = 1 TO 31: A$(A) = SPRITE$(63+A): NEXT: RETURN
340 FOR A = 1 TO 30: A$(A) = RIGHT$(SPRITE$(63+A),30)+LEFT$
(SPRITE$(A+64),2): NEXT: RETURN
350 FOR A = 1 TO 30: SPRITE$(63+A) = A$(A): NEXT: RETURN
360 CX = PEEK (-1532):CY = PEEK (-1533): PRINT O$;:RETURN
370 POKE -1532,CX: POKE -1533,CY: PRINT Z$;:RETURN
380 FOR A = 0 TO 920 STEP 40: VPOKE A + OS, 0: NEXT: RETURN
390 GOSUB 300: LOCATE 2,0: F$="": PRINT "FILENAME TO "OP$">";:
LINEINPUT F$: GOSUB 310
400 IF INSTR(F$,"cas:")=1 THEN MID$(F$,1,4)="CAS:"
410 RETURN
420 MOTOR OFF: IF INKEY$<>CHR$(13) GOTO 420 ELSE RETURN
430 DEFUSR = &H3498: J=USR(0): SCREEN,1: LOCATE,1:END
440 GOSUB 300: LOCATE1,0: PRINT "SET TAPE TO CORRECT LOCATION":
PRINT " PRESS SPACEBAR WHEN READY"
450 IF STRIG (0) = 0 GOTO 450
460 LOCATE 1,0: PRINT "PREPARE CASSETTE DECK"
OP$ = "SAVE" THEN PRINT "PRESS PLAY & RECORD"
PRINT " PRESS PLAY"
470 PRINT " PRESS ENTER WHEN DONE": GOSUB 420: GOSUB 310
480 GOSUB 360: POKE -1533,25: RETURN

```

----- THE GREAT 'SWEEP' COMPETITION

Those of you who operate CP/M will know of a great Public Domain Program called 'SWEEP' (or 'NSWP' or 'NEW SWEEP'). which is probably the second most used program available (The first being Good Ole Wordstar).

What is SWEEP?

For those of you who are not using CP/M, or are novices in the SV world, SWEEP is a file-handling Utility. It can;-

Copy Rename Delete Squeeze/Unsqueeze (ie 'pack')
and manipulate in many other ways, any form of CP/M file.

All This In one Handy Package.

UNFORTUNATELY THERE IS NO EQUIVALENT FOR BASIC DISK FILES.

The Committee therefore proposes a competition to produce a suitable Utility which will perform a wide variety of File handling Functions.

The competition is not however limited to a SWEEP look-alike search.

Any Utility program will be considered for this competition.

As prizes, the committee has bought a box or two of Blank Disks - more grist for the mill.

As the objective of the competition is to get more Public Domain Software, all entries will be accepted on the basis that they may be included in the Public Domain.

HOWEVER, If you really think that your Brainchild is the best thing around, ... send it to us for evaluation (Telling us NOT to put it in Public Domain) and we will review it for you.

As an indication of the sort of thing we are after, Jim Oliver has submitted the following program :

SWP23, A BASIC VERSION OF "SWEEP"

SWP23 was devised to offer a more convenient way to copy, kill, rename, print, view etc, any Basic disc file.

This version is offered for use and modification in the hope that you, the reader will respond with your ideas to improve on it and perhaps come up with something more useful and idiot proof.

Anyone familiar with the CP/M files "SWEEP" or "NSWEEP" will need no tuition in the use of SWP23. Anyone not using the "SWEEPS"; should!

When this programme is run it reads the directory tracks and fills an array with the file names. (This part of SWP23 is from a recent Menu Demo programme in the Waikato Newsletter by Brian Dorofaeff.) The programme then prints out various disc parameters and the first file name in the directory then waits for input at line 230.

Pressing ENTER or the SPACE BAR will move the cursor on though the file directory until the end where it will start from number one again.

The program waits at line 230 for you to input a single letter command. A list of the commands is shown at the end of the listing.

Where relevant, the command you input acts only on the file against which the cursor is sitting at the time of inputting the command.

One of the most useful things this program does is enable you to "TAG" a number of files for copying to the other (or same) disc. TAGGING thus avoids a lot of extra typing. The tagging process is remembered by the program and the already tagged and copied files can be RETAGGED or UNTAGGED depending on your whim. Tagged but used files are shown with a # after them. The A, M, or U commands when entered at the end of a file name on the screen will only act on TAGGED files, so remember to TAG first before using these.

The more astute user will note that certain commands require the effort of the ENTER key after the input of the command. This is to force you to think a little about whether to kill or copy or not. It takes a while to get used to which are which in this regard but this can be worth the effort when a valuable file is about to be lost.

I would'nt say that the program is free from bugs, on the contrary, I know it's got a few. But please, let us know what you think.

Remember that once the program is in memory, discs can be changed at will and operated on continuously without reloading the program. This technique is unusual at first but very useful, (particularly in CP/M under the "SWEEPS".)

```

10 REM SPECTRAVIDEO BASIC SWEEP
20 REM J.A.OLIVER MARCH 1987
30 MAXFILES=2
40 CLEAR 800
50 DIM FL$(79)
60 IV$=CHR$(27)+"p":NV$=CHR$(27)+"q"
70 CM$="AaBbCcDdFfLIIMpPrRsSsTtUuVv"+CHR$(32)+CHR$(13)+"Xx?"
80 SCREEN,0:CLS:CLOSE
90 PRINT"Log in;";IV$+" which drive 1 or 2 "+NV$;" ";:
D$=INPUT$(1):PRINTD$
100 'read files
110 D=VAL(D$)
120 FOR SEC=1 TO 14:C$=DSK1$(D,20,SEC)
130 FOR I=1 TO 16:A=ASC(MID$(C$,I*16-15)):IF A=0 THEN 170
140 IF A=255 THEN 180
150 NF=NF+1
160 FL$(NF)=MID$(C$,I*16-15,9)
170 NEXT:I
180 IF NF<0 THEN END
190 PRINT:PRINT "Number of files on disc: ";NF:PRINT
200 PRINT USING "###.##"; DSKF(D)*4.0961;:PRINT" K bytes left on
drive ";D:PRINT:PRINTDSKF(D);" tracks free."
210 RE=1:PRINT:PRINT
220 PRINTD$+" "+FL$(RE);
230 IN$=INPUT$(1):PRINTIN$
240 C=INSTR(CM$,IN$)
250 ON C GOTO
290,290,350,350,390,390,440,440,500,500,570,570,590,590,700,700,
770,770,810,810,840,840,880,880,940,940,260,260,1000,1000,1030

```

```

260 REM Repeat for next file
270 IF RE=NF THEN 210
280 RE=RE+1:PRINTD$+": "+FL$(RE);:GOTO 230
290 REM Retag files
300 FOR T=1 TO NF
310 IF LEN(FL$(T))=9 THEN 330
320 IF MID$(FL$(T),10)="#" THEN FL$(T)=LEFT$(FL$(T),9):
FL$(T)=FL$(T)+"t":PRINT"Retagging ";D$+": "+FL$(T)
330 NEXT T
340 GOTO 220
350 REM (B)ack for previous file
360 IF RE=1 THEN RE=NF:PRINT:GOTO 220
370 RE=RE-1:PRINTD$+": "+FL$(RE);
380 IF RE<0 THEN 210 ELSE 230
390 REM Copy files
400 PRINT: INPUT"Copy TO which drive, 1 or 2";CD$
410 PRINT:PRINT"copying "D$+": "+FL$(RE)+"====> "+CD$+": "
+FL$(RE)
420 COPY VAL(CD$)FROM D$+": "+FL$(RE)
430 GOTO 220
440 REM Kill files (permanently)
450 INPUT"Delete file y/n";KF$
460 ON INSTR("YyNn",KF$)GOTO 480,480,220,220
470 GOTO 220
480 KILL D$+": "+FL$(RE):FL$(RE)=""
490 IF FL$(RE)="" THEN FL$(RE)="Now dead!":GOTO220
500 REM Find files
510 INPUT "Find what";IN$:FLAG=0:RE=1
520 IF RE=NF THEN PRINT"Can't find it, try again":PRINT:GOTO 220
530 REM This line is not missing!
540 IF IN$=LEFT$(FL$(RE),LEN(IN$)) THEN PRINTD$+": "
+FL$(RE);:FLAG=1
550 IF FLAG=0 THEN RE=RE+1:GOTO 530
560 GOTO 230
570 REM Log in new drive
580 CLS:RUN 30
590 REM Mass copy
600 INPUT"Copy to which drive, 1 or 2";CD$
610 FOR T=1 TO NF
620 IF LEN(FL$(T))=9 THEN 680
630 IF MID$(FL$(T),10,1)="#" THEN FL$(T)=LEFT$(FL$(T),9)
640 PRINT:PRINT"copying "D$+": "+FL$(T)+"====> "+CD$+": "+FL$(T)
650 COPY VAL(CD$) FROM D$+": "+FL$(T)
660 DW$=LEFT$(FL$(T),9)
670 FL$(T)=DW$+"#"
680 NEXT T
690 GOTO 220

```

```

700 REM Print files
710 PRINTTAB(6)IV$+" Files must be saved as ASCII "+NV$:PRINT
:PRINTTAB(8)"press any key to continue":PRINT:PRINTTAB(12)IV$
+"esc"+NV$+" to return":RE$=INPUT$(1):IF RE$=CHR$(27) THEN 220
720 OPEN "LPT:" FOR OUTPUT AS #1
730 OPEN D$+" "+FL$(RE) FOR INPUT AS #2
740 INPUT #2,A$:PRINT #1,A$:A$=""
750 IF EOF(2) THEN CLOSE ELSE 740
760 GOTO 220
770 REM New name
780 INPUT "New name";NN$
790 NAME D$+" "+FL$(RE) AS D$+" "+NN$
800 FL$(RE)=NN$:GOTO 220
810 REM Remaining space
820 PRINT "Which drive: ";ND$=INPUT$(1):PRINTND$:ND=VAL(ND$)
830 PRINT USING "###.##";DSKF(ND)*4.0961::PRINT "K bytes remain":
GOTO 220
840 REM Tag files
850 FL$(RE)=FL$(RE)+"t":IF LEN(FL$(RE)) >10 THEN FL$(RE)=LEFT$
(FL$(RE),10)
860 RE=RE+1:IF RE=NF THEN 210
870 GOTO 220
880 REM Untag files
890 FOR T=1 TO NF
900 IF LEN(FL$(T))=9 THEN 920
910 IF MID$(FL$(T),10)="t" OR MID$(FL$(T),10)="*" THEN FL$(T)=
LEFT$(FL$(T),9):PRINT "Untagging ";D$+" "+FL$(T)
920 NEXT T
930 GOTO 220
940 REM View file
950 PRINTTAB(6)IV$+" Files must be saved as ASCII "+NV$:PRINT:
PRINTTAB(8)"press any key to continue":PRINT:PRINTTAB(12)IV$+
"esc"+NV$+" to return":RE$=INPUT$(1):IF RE$=CHR$(27) THEN 220
960 OPEN D$+" "+FL$(RE)FOR INPUT AS #1
970 INPUT #1,A$:PRINTA$:A$=""
980 IF EOF(1) THEN 990 ELSE 970
990 CLOSE:GOTO 220
1000 CLOSE:FILES D: SCREEN,1:CLS:LOCATE 9,10:PRINTIV$+" Please
check files ":LOCATE 9,11:PRINT " before switching off "+NV$:
PRINT:PRINTTAB(3)"If you get a Bad Allocation Table":PRINT
:PRINTTAB(3)"try typing CLOSE:FILES and <enter>."
1010 REM if that doesn't work don't ring me!
1020 END
1030 LOAD "1:screen":LOCATE1,18:PRINTSTRING$(38," "):LOCATE2,21:
PRINTIV$+" Run swp23 y/n "+NV$:R$=INPUT$(1):IF R$="y" OR R$="Y"
THEN RUN 40
1040 SCREEN,1:END

```

swp23 HELP INSTRUCTIONS

A = RETAG FILE
B = BACK ONE FILE
C = COPY FILE
D = DELETE FILE
F = FIND FILE
L = LOG IN NEW DISC
M = MASS COPY TAGGED FILES
P = PRINT ASCII SAVED FILE
R = RENAME FILE
S = CHECK DISC SPACE
T = TAG FILES
U = UNTAG FILES
V = VIEW FILES ON SCREEN
X = EXIT THIS PROGRAM
? = DISPLAY THIS HELP

I edited the above helps onto a blank screen using the normal command level editor, then saved it as a screen image using SAVE "1:screen",S.

This can then be called onto the screen in line 1030 without affecting the file already in memory. You are now prompted to either run or not. The printing of string\$(38," ") just rubs out the "save "1:screen" statement which is unavoidable if you want to save a screen image. You may need to alter the LOCATE statment to achieve this depending on your exact screen. The screen image can only be saved from an actual screen image, so you can't copy it directly from one disc to another. Just LOAD the image to the screen then SAVE it to disc to create another copy. This is done with a two line program.

NOTE: The S.V. COPY command has a bug in it somewhere which causes it to occasionally "under copy" some files. It is as well to check the size of the copied file (in tracks) using CLOSE:FILES after you have finished with swp23.

Jim Oliver

Tricks with WORDSTAR and OWIKKEY

by Graeme Dick

The following ideas are not original, having been picked up from computer magazines. You may find them useful.

WORDSTAR

Heading and Footings

If you have had occasion to use this Wordstar feature - the .HE and .FO dot commands - you will know that you are limited to a one-line message in each case. You might think it would look nice to separate the heading/footing from the rest of the page - for example, something like this:

This is a 2-line footing which prints at the bottom of each page

The trick is to make use of a characteristic of most printers of automatically doing a 'carriage return + linefeed' when the maximum number of characters for the line has been reached.

If your printer is set up to do Pica type (10 characters/inch, i.e. 80/line), the <cr><lf> will occur after 80 characters. To get a two-line footing, say, you turn off the wordwrap feature (^OW) and set up an extra long line for the .FO command - allowance must be made for the page offset (the left margin, set by the .PO command, which defaults to 8 characters in Wordstar), and for the characters not used from your maximum line length.

For Pica type, with the default values, you will have:

```
65      first line message. Default line length = 65
7        spaces at end of first line [80 - (65+8)]. Page
        offset = 8
8        spaces for page offset for second line - must be
        entered
65      second line message
--
145
```

You could even go on to a third line by adding another 15 spaces (7+8) then the message. However, Wordstar will only print something like 194 characters on a line even though you can have up to, I think, 240 on screen.

It is easier to set up if the dot command is added after the text of the heading/footing is entered. Having set this up with Wordwrap off (to prevent the line reforming), then turn Wordwrap back on for the rest of your article. The extra long line will be shown on screen by the + flag in the right-most column.

This would be the setup for two lines with the first line being a dashed line.

55 dashes

.FO----- (15 spaces) This is a two-line message etc
The one other thing you must do is decrease the page length by 1 for each extra line of the heading/footer. This is necessary because Wordstar doesn't know anything about the extra line printed and would get out of step.

To summarise:

1. Turn off Wordwrap
2. Set up heading or footing as described
3. Add dot command
4. Adjust page length using the .PL command - default is .PL66
5. Turn on Wordwrap
6. If you reformat your article, make sure that the long heading/footer is not reformatted.

Things to do with QwikKey

Notes have been provided by the author of the programme, and these are self-explanatory. Here are a few things that you can do with the programme.

I have set up the key definition file, here called KEYDEF.DAT, so that QwikKey uses the 'backslash' key (i.e. the Integer division key) to initiate and terminate definition of a key. I couldn't print that character here because QwikKey was loaded. It is a rarely used key nevertheless, but you can define your own choice of initiator - you could, for example, use one of the LeftGraph or RightGraph keys. This is probably a better idea, as you remove the risk of inadvertently hitting the initiator key.

I have defined the following keys in the KEYDEF.DAT file:

LeftGraph B = reform paragraph and return cursor to current place (Wordstar)

^B ^OP if you were typing the individual commands.

LeftGraph K = Save Wordstar file & return cursor to current place

^KS ^OP

LeftGraph O = [chr\$(96)]

LeftGraph P = | [chr\$(124)]

LeftGraph S = a combination of keystrokes to save a file in Supercalc, creating a backup copy and saving All the file. To enter each command individually requires /S ^M ESC ^M B A where ^M is the code for <enter>

LeftGraph W = delete word left (Wordstar) - it uses a combination of keystrokes to achieve this:
KK "A "H KE KY "B "OP

You can have fun creating your own code combinations.

The key definition files can be renamed to your convenience, e.g. you may wish to have a file SC2.KEY for Supercalc, WS.KEY for Wordstar, etc.

One of the big advantages of QwikkKey is that, once loaded, it can be used from within programmes such as Wordstar, Supercalc (and presumably DBASE) to create key combinations 'on the fly' - i.e. as you need them. These combinations can be saved if you desire, or lost when you quit.

The KEYFILES are saved using the entry format: OK S filename.typ

The drive can be specified in front of the filename in the usual way.

The QwikkKey author states that the maximum length of a key definition is 32 characters. I have found that, if you have only a few keys defined, the length can be considerably greater than this.

QwikkKey does seem to hang up occasionally - notably when trying to use STAT and some of the in-built functions. This is a minor inconvenience compared with the benefits that the programme has. The hangups don't seem to occur from within programmes, or at least not to the extent of meaning that you will lose your file - it is nevertheless good practice at any time to do frequent saves of your work.

I have noticed with Supercalc that when memory is getting full, the QwikkKey programme does interfere by preventing entry of any more data even though there should be room. You'll have to keep an eye out for this sort of thing.

Graeme Dick

SV Graphics Tablet

There is not a great deal of information on the SV Graphics Tablet or any graphics tablet for that matter. The next three programs may be interesting enough to stimulate some new ideas hopefully more inspired!

The first one (lines 10-80) illustrates the use of basic to allow the tablet (pad) to input some numbers which are used to create a series of musical notes. The scaling formulae are essential to make the physical lengths on the pad surface more closely match the SOUND requirement.

The second program (100-170) is a simple way to print garbage on the screen (or printer if so desired). By predefining strings (your name, say) it is possible to have these print onto the screen when a certain physical point is touched on the pad. The screen colour may be changed by touching a different point and so on.

The last and longest program was designed to create sprites "by hand" on the pad. The screen starts off with a white area in the top LH corner. By touching the pad this area can be erased in any pattern desired but within the rather poor definition. This area is located in a similar position on the pad.

By pressing "I", when you feel you have finished creating your masterpiece, the computer will then load this pattern and display it on SCREEN 1. The sprite can then be viewed whisking around the screen in response to your movements over the pad.

By putting in a bit of simple code after line 440 it is possible to SAVE the sprite pattern on tape or disc for later use.

Using this technique is not easy at first but with practice it becomes an easy way to create large sprites.

```

10 REM **** use pad for this one ****
20 DEFNG A-Z
30 IF PAD(0)=0 THEN 30 'is pad touched?
40 X=PAD(1):Y=PAD(2)
50 Z=X/10:A=ABS(A):A=LOG(Z):B=Y*1.3 'scale axes
60 SOUND 7,254:SOUND8,16:SOUND1,A:SOUND0,B:SOUND13,0:SOUND12,5
70 GOTO 30
80 END
100 REM *** pad printer ***
110 DEFINT C,D:SCREEN,0:CLS:X=0
120 ON STOP GOSUB 170:STOP ON
130 IF PAD(0)=0 THEN 130
140 A=PAD(1):B=PAD(2):C=A/6:D=B/8
150 LOCATE C,D:PRINT CHR$(A)
160 GOTO 130
170 CLS:SCREEN,1:END
200 REM pad sprite designer/Jim Oliver
210 CLS:D=0:B$="": '10/86
220 GOSUB 560: 'mark out work area
230 REM pad curser and rub out routine
240 ON INTERVAL=50GOSUB320: 'auto jump
250 INTERVAL OFF: 'stop jumping
260 IF D=257 THEN GOTO 470 'show off
270 INTERVAL ON: 'start jumping again
280 IF PAD(0)=0 THEN 280: 'stand still
290 X=PAD(1):Y=PAD(2):X=X/6:Y=Y/8: 'scale pad axes
300 LOCATE X,Y:PRINT CHR$(201): 'print
310 GOTO 280 'pad still active?

```

```

320 REM read vram and build sprite
330 I$=INKEY$:IF I$="" THEN RETURN: 'I'm still playing
340 IF I$<>"I" THEN RETURN: 'I for "load"
350 B=42:C=B+7:LOCATE 12,20:PRINT"Loading..."
360 C=B+7: 'adjust the count
370 FOR A=B TO C: '1st line, 8 bytes
380 D=D+1: 'flag
390 P=VPEEK(A):IF P=233 THEN A$="1" ELSE A$="0": 'binary
400 B$=B$+A$: 'make bin'y strings
410 IF D=129 THEN B=50:GOTO 360: 'first
420 NEXT A: '16 X 8 array done
430 A$="": 'new string please
440 C$=C$+CHR$(VAL("&b"+B$)):B$=""
450 IF D=257 THEN RETURN 250: 'end 2nd lot
460 B=C+33:C=B+7:GOTO 370: 'next line
470 REM display away! 'of vram
480 ON STOP GOSUB 550:STOP ON
490 COLOR1,1:SCREEN1,3: 'set up screen
500 SPRITE$(1)=C$: 'we hope!
510 PUT SPRITE 0,(X,Y),11,1: 'yellow
520 X=PAD(1):Y=PAD(2): 'OK for screen1
530 IF PAD(0)=0 THEN 530: 'dont move unless
'Im touching the pad
'I might still be
moving
540 GOTO 510:

550 SCREEN 0:COLOR 15,4,5:END
560 'chr$(200) swapped for chr$(211)
570 FORA=41TO641STEP40
580 VPOKEA,(200+32)
590 NEXT A
600 FORA=2TO17
610 VPOKEA,(171+32)
620 NEXT A
630 FORA=58TO658STEP40
640 VPOKEA,(197+32)
650 NEXT A
660 FORA=697TO682STEP-1
670 VPOKEA,(174+32)
680 NEXT A
690 RETURN

```

Jim Oliver

PS. Don Stanley has produced a disc loading version of the SV Graphics Tablet software. For anyone with disc drives who uses the tape version this is a real boon. Please contact Jim Oliver who will provide a copy if you can show proof of purchase of the tape version.

=====

EXPLORING BASIC - Part 5

By L. A. Dunning
(Courtesy of SV Australasian Users Group)

This month, I'll describe SCREEN 1. Due to problems I'm unable to describe SCREEN 2 as promised. This will, however appear next month. It is more difficult to analyse the modes other than TEXT, however a general description can be given.

GRAPHICS II (Screen 1)

The display is in Graphics 2 mode when Bit 1 of register 0 is set to 1 and Bits 4 and 3 of Register 2 are set to 0. In this mode the screen is divided into 32 positions, each defining a block of 8 X 8 pixels giving a full resolution of 256 across by 192 down. Things are not however as simple as they seem.

The screen is divided into three smaller screens which for the sake of clarity I shall call "top", "middle", and "bottom". The top screen describes the top 8 rows of positions, the middle describes the next 8 rows and the bottom describes the last 8 rows. Each screen operates by using a third of the pattern Name Table (NTB starts at 1800H), Pattern Generator Table (PGB starts at 0000H), and the pattern Colour Table (COLB starts at 2000H). The top screen uses the initial third of these tables, the middle screen uses the second third of the tables and the bottom screen uses the last third.

Each portion of the NTB consists of 256 bytes and allows any of 256 different patterns to be displayed at each position. As there are three of these, there can be 768 unique patterns on the screen, one for each position if necessary.

Each portion of the PGB contains definitions for each of the possible 256 patterns for that part of the screen, each definition being 8 bytes in length, so each portion is 2048 bytes long. Unlike text mode, all bits are used for pattern definition so each set of 8 bytes produces an 8 X 8 pattern. Bits set to 1 indicate foreground colour is used, bits set to 0 indicate that background colour is used.

Lastly, each portion of the COLB contain codes for that portion of the screen. Each line of a pattern can have two colours, background and foreground. Thus the COLB is the same length as the PGB requiring 8 bytes for each position. The bits 0-3 define background colour the bits 4-7 define foreground colour.

When SCREEN 1 is set up, all bytes in the PGB are set to zero; a loop dumps the values 0,1,2,... ...,255; once for each third of the screen (this setup never changes while in this screen); all bytes in the COLB are set for Background/foreground values.

How does this tie together? The pattern name table and the pattern generator table have a one-to-one correspondence so that a value in the NTB indicates which 8-byte pattern will be displayed. This is like text mode, however each portion of the NTB/PGB is only relevant to that portion of the screen. Rather than change a value in the NTB to display a different pattern on the screen, the software drivers change the patterns in the PGB instead, this is exactly reverse to the system used while in TEXT mode. Also of course, each position has its own colour definitions; this works but does produce unexpected results when trying to put 3 or more colours within an 8 pixel line. Diagram 2 illustrates the relationship between NTB, PGB, & COLB. Diagram 3 illustrates how colour works for each position.

Diagram 1 shows the mapping in VRAM of NTB & PGB tables, it may prove useful when mapping a display. To further illustrate the relationship between the various tables in SCREEN 1 I've included listing 1, which demonstrates different functions.

DIAGRAM 2

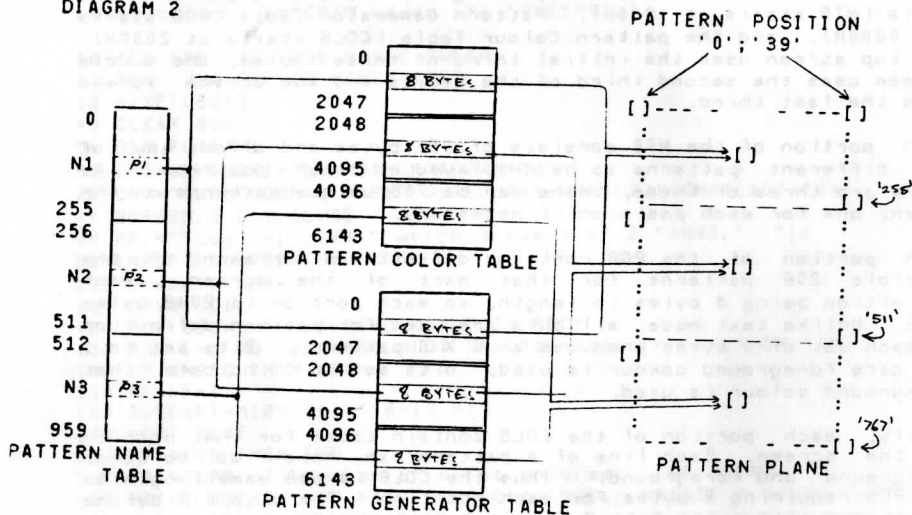
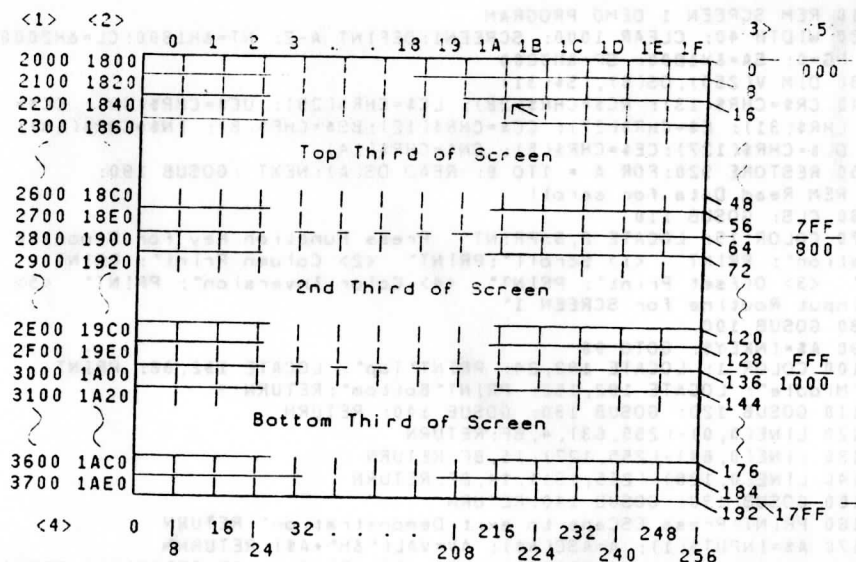


DIAGRAM 1



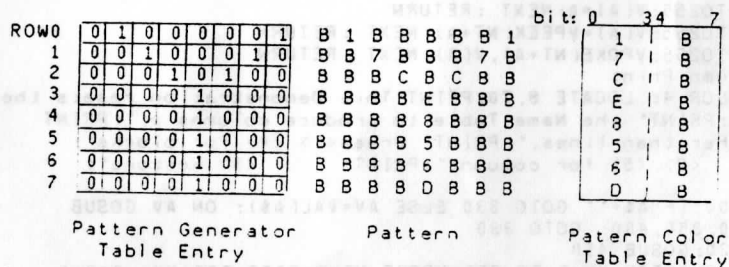
NOTES

- 1 COLOR TABLE ADDRESSES (Top Left Corner of nearest location)
- 2 PATTERN NAME ADDRESSES (ditto)
- 3 Pixel row referred to by 'Y Co-ordinate'
- 4 Pixel column referred to by 'X Co-ordinate'
- 5 PATTERN GENERATOR ADDRESSES (Top Left to Bottom Right)



Location by pixel ref (216,16)
by Pattern Name Table &H185B
by Color Table &H24C0

DIAGRAM 3



```

10 REM SCREEN 1 DEMO PROGRAM
20 WIDTH 40: CLEAR 1000: SCREEN1:DEFINT A-Z: NT=&H1800:CL=&H2000:
  PG=0: SA=&H1800: SP=&H3800
30 DIM V(255),DS(8),IS$(31)
40 CR$=CHR$(13): RC$=CHR$(28): LC$=CHR$(29): UC$=CHR$(30): DC$=
  CHR$(31): E$=CHR$(27): CC$=CHR$(12): BS$=CHR$(8): IN$=CHR$(18):
  DL$=CHR$(127): CE$=CHR$(5): CN$=CHR$(14)
50 RESTORE 920:FOR A = 1TO 8: READ DS(A):NEXT :GOSUB 190:
  REM Read Data for scroll
60 CLS: GOSUB 110
70 COLOR 15: LOCATE 0,8:PRINT" Press Function Key for Demonstr
  ation": PRINT" <1> Scroll":PRINT" <2> Column Print": PRINT
  " <3> Offset Print": PRINT" <4> Color Inversion": PRINT" <5>
  Input Routine for SCREEN 1"
80 GOSUB 100
90 A$=INKEY$: GOTO 90
100 COLOR 1: LOCATE 192,24: PRINT"Top": LOCATE 192,88: PRINT
  "Middle": LOCATE 192,152: PRINT"Bottom":RETURN
110 GOSUB 120: GOSUB 130: GOSUB 140: RETURN
120 LINE(0,0)-(255,63),4,BF:RETURN
130 LINE(0,64)-(255,127),15,BF:RETURN
140 LINE(0,128)-(255,191),14,BF:RETURN
150 GOSUB 130: GOSUB 140:RETURN
160 PRINT"Press ESCape to exit Demonstration":RETURN
170 A$=INPUT$(1): A=ASC(A$): AV=VAL("&H"+A$):RETURN
180 DD=STICK(0) OR STICK(1): ST=ABS(STRIG(0) OR STRIG(1)):RETURN
190 STOP ON: ON STOP GOSUB 930: KEY ON : ON KEY GOSUB 230,320,470
  ,600,650 : RETURN
200 PRINT:PRINT"Press <ENTER> to continue"
210 GOSUB 170: IF A$<>CR$ GOTO 210 ELSE RETURN
220 REM Scroll
230 DM =2: LINE(0,0)-(255,63),1,B: COLOR 4: LOCATE 8,74:PRINT
  "Use Cursor Keys or Joystick":PRINT"to Scroll Top Segment": PRINT
  "Press SPACEBAR/FIRE to restore": PRINT"All Scrolling is Circu
  lar":GOSUB 160
240 GOSUB 290
250 GOSUB 900: GOSUB 180: IF DD + ST = 0 GOTO 250
260 GOSUB 270: ON ST GOSUB 280:GOSUB 300: GOTO 240
270 FOR A=0TO255:V(A)=(V(A)+DS(DD))MOD 256:NEXT :RETURN
280 FOR A=0TO255:V(A)=A:NEXT :RETURN
290 FOR A=0TO255:V(A)=VPEEK(NT+A):NEXT :RETURN
300 FOR A=0TO255:VPOKE(NT+A),V(A):NEXT :RETURN
310 REM Column Print
320 DM=2:COLOR 4: LOCATE 8,70:PRINT"This Demonstration resets the
  values on":PRINT" the Name Table to produce columns of":PRINT
  " PRINT rather than lines":PRINT" Press<1> <4> for blocks":
  PRINT" <2> <5> for columns":PRINT" <3> Restore":
  GOSUB 160
330 GOSUB 900:IF A$="" GOTO 330 ELSE AV=VAL(A$): ON AV GOSUB
  340,390,410,350,400: GOTO 330
340 GOSUB 120:GOSUB 450
350 B=0:R=0:C=0:FOR XX=0 TO 255:VPOKE N1+C+R*32+B*6,XX: C=C+1
360 IF C>5 OR B*6+C>31 THEN C=0: R=R+1
370 IF R>7 THEN R=0: B=B+1
380 NEXT :RETURN

```

```

390 GOSUB 120:GOSUB 420
400 XX=0: FOR A=0TO31: FOR B=0TO7: VPOKE NT+B*32+A,XX:XX=XX+1:
NEXT B,A:RETURN
410 GOSUB 280:GOSUB 300:RETURN
420 X$="This is a test to demonstrate how to Change the SCREENS
' ' ' ' show columns. Notice how the <ENTER> message is confu
sed by the change.":GOSUB 430: GOSUB 200:RETURN
430 COLOR 15:X=2:Y=1:FOR N=1TO LEN(X$):LOCATE X*8,Y*8:PRINTMID$
(X$,N,1):LOCATE X*8,Y*8:PRINT"H":X=X+1:IF X >31 THEN X=0:Y=Y+1
440 NEXT :RETURN
450 LOCATE 0,0: COLOR 15:PRINT" YOUR MESSAGE,ANY MESSAGE (LENGT
H < 40)":PRINT:PRINT:PRINT"Substitute message in line 450 ":
PRINT:GOSUB 200:RETURN " * Insert message here, not more than
40""Start message at right location"
460 REM Offset Print
470 DM=5:COLOR 15,0,4:SCREEN 1: LOCATE 8,0:PRINT"Words are just
PIXELS, O.K."
480 DEF FNX = 250-X: DEF FNY = Y+9 : GOSUB 550
490 DEF FNX = 250-Y: DEF FNY = X : GOSUB 550
500 DEF FNX = 250-X: DEF FNY = 180-Y : GOSUB 550
510 DEF FNX = Y+20 : DEF FNY = 199-X : GOSUB 550
520 DEF FNX = 16+X-Y : DEF FNY = 9+Y+X : GOSUB 550
530 DEF FNX = X*2 : DEF FNY = 50+Y*2 : GOSUB 550
540 GOSUB 900:GOSUB 540
550 FOR X = 0 TO 175: FOR Y = 0TO7
560 DO = POINT(X,Y): IF DO = 0 GOTO 580
570 PSET (FNX,FNY),DO
580 NEXT Y,X:RETURN
590 REM Color Inversion
600 DM = 5: LOCATE 8,80:PRINT"Which Screen to invert":COLOR 4:
PRINT" <1> Top": COLOR 1: PRINT" <2> Middle": COLOR 14: PRINT
" <3> Bottom": COLOR 15: LOCATE 8,152: PRINT " Hmmm...?":
COLOR 1: GOSUB 160
610 GOSUB 170: IF A$=E$ THEN GOSUB 910
620 IF AV <1 OR AV >3 GOTO 610 ELSE TZ = (AV-1)*2048+CL
630 FOR I = TZ TO TZ+2047: J= VPEEK(I): MS = J\16: LS = J MOD 16:
JN = MS+LS*16: VPOKE I,JN:NEXT :GOTO 610
640 REM Input$ Routine
650 DM = 4: LOCATE 8,68: COLOR 4: PRINT"This demonstration uses
an input routine":PRINT" for SCREEN 1. It follows most rules
of LINEINPUT":PRINT" For an explanation, list lines:"
660 PRINT" lines 640 - 840": GOSUB 160
670 XX=1: YY=17: LL=1:IM=0:SL=1:ML=29:FG=1:BG=14:GOSUB 850:
X1=XX*8: Y1=YY*8: COLOR FG,BG:LS$="": GOSUB 880
680 PUT SPRITE0,(X1+SL*8-1,Y1-2),15,ABS(IM)
690 A$=INPUT$(1): IF A$ = CR$ GOTO 820
700 IF A$ = E$ THEN GOSUB 910
710 IF A$ = CE$ AND LL>0 THEN LL = SL: GOSUB 880
720 IF A$ = CN$ AND LL>0 THEN SL = LL
730 IF A$ = LC$ THEN SL= SL+(SL>1): IM =0
740 IF A$ = RC$ THEN SL= SL-(SL<ML): IM =0
750 IF A$ = BS$ AND SL>1 THEN SL= SL-1 :GOSUB 860
760 IF A$ = IN$ THEN IM = NOT IM
770 IF A$ = DL$ AND LL>SL THEN GOSUB 860: GOTO 680
780 IF A$<" " GOTO 680

```

```

790 IF IM THEN GOSUB 890
800 IS$(SL) = A$: A= SL: GOSUB 870: SL=SL-(SL<ML): IF LL<SL THEN
  LL = SL-1
810 GOTO 680
820 IF LL THEN FOR A = 1 TO LL: IS$=IS$+IS$(A):NEXT
830 LINE (5,160)-(255,191),BG,BF: LOCATE 8,160:PRINT
  "The string is:":PRINT " >"IS$"< ":PRINT "Press SPACEBAR for
  another example"
840 GOSUB 170: IF A$=E$ THEN GOSUB 910 ELSE IF A$<>" " GOTO 840
  ELSE GOTO 670
850 SCREEN ,0:SPRITE$(0)=CHR$(255)+STRING$(7,129): SPRITE$(1)=
  STRING$(3,0)+CHR$(255)+STRING$(3,129)+CHR$(255): RETURN
860 FOR A = SL TO 30: IS$(A) = IS$(A+1): GOSUB 870:NEXT : LL=LL-1
:RETURN
870 LINE (X1+A*8,Y1-1)-(X1+A*8+7,Y1+9),BG,BF: LOCATE X1+A*8,Y1:
  PRINTIS$(A):RETURN
880 FOR A = SL TO 30:IS$(A) = " ": GOSUB 870: NEXT :LL=LL-1 :
  RETURN
890 FOR A = 30 TO SL+1 STEP-1: IS$(A)=IS$(A-1): GOSUB 870: NEXT:
  RETURN
900 A$=INKEY$: IF A$<>E$ THEN RETURN
910 GOSUB 850:GOSUB 410: ON DM GOSUB 120,130,140,150,110:
  GOSUB 120:GOSUB 100:GOSUB 190: RETURN 70
920 DATA 32,31,255,223,224,225,257,33
930 COLOR 15,4,1

```

 From the oops - you blew it editor department, I must confess that I did not re-read these articles ahead for typo's and error corrections - so please note the following correction

The table in Part 4 referring to the 'COLB' pointer in Register 3 should refer to VRAM location &H2000 (Not 3FC0) HOWEVER - I started investigating the values to be output to the registers. I found that outputting '&HFF' to Register 3 set the color table OK - this shouldn't have worked !! as &HFF indicates VRAM location &H3FC0. Further investigation showed that the Pattern Generator Table value '&H00' was in the same way corrupted. The Pattern Generator Table does start at location &H0000 in VRAM, however to set up this value you have to pass '&H03' to Register 2.

I can't explain the above errors and would therefore welcome some ideas - as I hate being beaten by something like that !!

Also I haven't checked whether any of the listings (in previous issues - EXPLORING BASIC SERIES) run as printed - so what I will do is type the next listing in under BASIC and try to check that it runs (time permitting). This will have the added advantage that I can pass the listings on to Jim for the Public Domain. (All the previous listings will go across in the same way sometime soon.)

Steve Hanson

TITBITS

THE PROBLEM:

THE EXPLANATION

This uses a form of the COPY command that I only found out about quite recently.

The COPY command is usually used in the format;

COPY (DN) FROM (DN) - Where (DN) is 1 or 2 for the Drive Number

The other options :-

COPY (DN) FROM "D:F-NAME.EXT" - Copies the file "F-NAME.EXT" from drive D: to drive (DN).

COPY D:(TD,SD) FROM DN:(TSS,SSS)-(TSE,SSE) - Copies physical sectors from drive DN; to drive D;. The Source of the copy will Start from Track TSS, Sector SSS. The sector copy will End at Source Disk Track TSE/Sector SSE. The Destination Disk will receive the Copy commencing from Track TD, Sector SD.

As for the error described above, when copying a file using the COPY command, all File Naming Conventions Must Be followed. If the file name does not contain a device descriptor, ie 'CAS:' for Cassette files; '1:' or '2:' for disk files, the default of cassette is assumed (By the routine in ROM which validates filenames). The program segment above does not give the correct device descriptor, as it does not include a colon (:) after the drive number. The ROM routine, therefore starts looking for a cassette file called "E"+A\$, rather than a file called A\$ on drive "E:". Hence the Tape message. The program should read:-

```
10 E$="2":A$="ipl"
```

```
20 COPY 1 FROM E$+"":A$
```

(Ed,)

STOP PRESS !

MY
CO-EDITOR

HAS JUST BECOME
A DAD !

congratulations! Joe,
from us all

(Ed:)

28 Warden Street,
Opoho,
Dunedin.
17 February 1987

The Secretary,
Wellington User Group,
P.O. Box 26050,
Newlands,
Wellington.

Dear Sir,

I am writing to advise that for the interest of any of your group members, I have items of Spectravideo hardware and software available for purchase. All items are in excellent condition.

I set out a list below of all the items involved. While I would prefer to sell as a package deal, individual items would be available if required. At this stage I leave all items open to reasonable offers, with the right of refusal if offers are totally unacceptable. The software would be included with the tape cassette drive or disc expander accordingly, and the list is only to indicate what is included. I do not intend to sell the software or manuals on their own. Any interested member may contact me at the above address.

ITEMS AVAILABLE:-

SV328 computer with SV903 tape cassette drive. Computer has Reset switch fitted.

SV6058M Expander. This incorporates dual DSDD drives giving 640K storage capacity, 80 Column card with front panel switch for 80/40 column display selection, Centronics interface for printer (cable included), composite video output & audio.

Phoenix colour monitor, RGB, composite video colour or black & white or green screen.

SV3300 15" dot-matrix printer, near letter quality or draft copy, 136 or 233 columns per line, 130 characters per second, IBM graphic character set, and bit image graphic capabilities.

SOFTWARE

WORDSTAR, DATASTAR, CALCSTAR, DATASTAR

All the above are complete with full original manuals giving potential for business use.

System Master Discs: CPM2.23, CFM2.22, SV Disc Basic plus other miscellaneous language and system discs.

Tapes: Introduction To Basic, Ninja, Spectron, Telebunny, Ferilous Journey, SVI memory expansion, SVI Magic Of Spectra-

video - with assembler and manual.

DISC SOFTWARE: Flight Simulator, and general games etc.

BOOKS: Basic Reference Manual, Basic on the Spectravideo, Graphics Sound & Sprites on the Spectravideo, plus CFM and users manuals supplied with hardware.

Other miscellaneous items such as a cartridge version of the MSX operating system for the SV328 and utilities etc would be included.

As stated above, any reasonable offer for the items will be considered. I would like to offer my best wishes to your group for the coming year.

Yours faithfully,

C. A. Adams.

C. A. Adams

SPECTRAVIDEO

USERS GROUP WELLINGTON

52700 = X * 26

X

652
CORRECTION: SCHOOL HOLIDAYS MEANS
MAY MEETING WILL BE *** 25 MAY ***
- AND TO JIM'S SWP23 PROGRAM LINES:
510 INPUT "Find What?"; IN\$;
IF FLAG = 0 THEN RE = 1
550 ..for 'GOTO 530' read 'GOTO 520'
560 ..change to 'FLAG=0:GOTO 230'